Tactix: restoration of the LymphoSAT SAT-constructor surface

Engineering checkpoint for Andrew

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This checkpoint restores the retained LymphoSAT-derived SAT constructors as ordinary Tactix actions. Every stable Lucky identity from 0 through 24 is again enumerable and replayable. The change is deliberately a SAT-recovery step: it does not turn bounded constructor failure into UNSAT, and it does not replace later certified theory-to-Boolean work for reusable deductions or refutations.

No benchmark, baseline, or portfolio experiment was launched. The expected owner recovery is based on previously checked saved evidence and restoration of the same retained implementations. Current target closure and performance remain unmeasured until the authorized regression and owner-instance gates are run.

Action and soundness boundary

Lucky IDs 6–14, 17, 18, and 21 retain their historical SAT-constructor meanings. IDs 15/16 remain proof-producing st-connectivity and signed cardinality actions; 19/20/22 remain startup proof actions; and 23/24 remain the separate proof-only parity and exact-one actions. In particular, IDs 6 and 7 no longer call the refutation engines. An inconsistent parity or exact-one structure therefore makes the SAT constructor decline, while the explicit proof identity may return UNSAT only through its certified path. The instance dispatcher also keeps SAT-only IDs ahead of the shared root propagation drain. Thus an unpropagated root contradiction cannot be misattributed to a constructor; direct ID-6/7 controls verify that they return no change without marking the kernel inconsistent. Proof-capable identities and the canonical probe cascade retain the root-closure path.

Every completed constructor assignment is checked against the whole live CNF before Tactix returns SAT. Parity, one-hot GF(3), affine, DirectCSP, Oddball, and graph-labeling searches can publish a signed partial assignment when they do not finish. A zero entry remains unknown; it is not converted to a false phase. Partial candidates affect only the shared guidance channel and cannot return SAT or UNSAT. Focused tests add variables outside the recovered structure and verify that they stay unknown in parity, one-hot, affine, DirectCSP, Oddball, and graph-labeling publications.

DefaultPolicy restores the historically enabled 6/7/9, 11–14, 17, and 18 opportunities and retains the restored Oddball ID 10 opportunity. Graph labeling ID 21 remains opt-in because its production effort dose has not been accepted. Affine ID 8 is public and random-selectable, but no historical DefaultPolicy stage existed, so this checkpoint does not invent one.

The random controller enumerates the entire finite Lucky catalog and binds each typed reference profile to the selected stable identity. The legacy random-program generator also binds sampled parity, one-hot, affine, Fermat, lookup, SIMON, st-connectivity, signed-cardinality, exact-cover, independent-set, graph, and Ramsey parameter types only to compatible actions. One-hot sampling selects either GF(3) ID 7 or DirectCSP ID 9. Configuration fingerprints, C configuration, C++ replay, feedback, and invalid

cross-identity rejection cover the restored typed paths.

Saved owner obligations

The table records prior independently checked evidence used to identify the recovery obligations. It is provenance for this implementation choice, not a new current-source solve claim.

Lucky ID	Mechanism	Saved checked owner evidence
6	parity SAT model	WV ranks 7 and 16, two-trees family
7	one-hot GF(3)	twelve WV100 `gf3-sat` owner rows
8	affine model	WV ranks 51, 63, 64, and 65
9	DirectCSP	WV ranks 93 and 98
10	Oddball balanced signature	ten direct checked models; Oddball 112 used candidate guidance followed by Walk/CDCL
11	Fermat circuit	WV rank 27
12	Černý reset word	WV rank 40
13	lookup circuit	WV rank 73
14	SIMON32 key search	WV rank 85
17	exact cover	WV2 `sgen1-sat-180-100`
18	independent set	WV2 `1-ET-512-K-96`
21	graph labeling	Tier-D rank 991, checked against all 178,213 original clauses

The saved sources are the LymphoSAT model-constructor report, the integration attribution/action-map reports, the Oddball AM161/AM333 artifacts, and the rank-991 graph target report already linked from `PROGRESS.md`. The current checkpoint does not claim that every row has been re-solved by this worktree.

Validation recorded at this checkpoint

Check	Result
Constructor, public API/replay, controller, configuration, feedback, and guidance selection	18/18 passed
Unknown-preserving parity, one-hot, and affine partial-candidate checks	3/3 passed
Additional policy/feedback integration selection	12/12 passed in 198.04 seconds
ASan/UBSan parity, one-hot, affine, public API and random replay	5/5 passed
Normal-duration broad `lucky` integration test	passed in 942.13 seconds
Documentation proof-rule, citation, and section-reference checks	passed
Documentation size check	one pre-existing failure: `tools/fuzz/README.md`, 91/80 lines

The fast set includes parity, one-hot, affine, Oddball, Fermat, Černý, lookup, SIMON, exact-cover, independent-set and graph model tests plus public C action enumeration/configuration, C++ replay,

both random selectors, DefaultStrategyConfig, startup catalog stability and local-search guidance. The checks verify all active IDs 0–24, exact typed identities, wrong-ID rejection, complete-model validation, the 6/7 versus 23/24 semantic split, and graph default-off behavior.

Limits and next actions

This source remains unaccepted. The B28+C5 and Rank1–50 regression gates are mandatory. WV/WV2 handling remains split: LymphoSAT-owned SAT rows must be recovered, while non-owner rows are preservation controls. Both-policy fuzz, N1–N4 review, independent original-CNF checking of current owner solves, and the accepted 30-minute proof-checking cap with the baseline plus $\max(30\%, 20\text{s})$ slowdown allowance still apply.

GF(3) UNSAT remains deferred until SAT recovery closes. Its next module should perform Gaussian elimination in a separate theory component and return exact premises, consequences or conflict, and a compact derivation object. The SAT adapter should validate and serialize DSR/DRAT proof steps without repeating the algebra or eagerly Booleanizing the entire system. No new PB path is planned unless measured clausal certification is first shown impractical. Lookup-circuit and other exact constructors may later expose coherent partial guidance, but that is separate from restoring their already checked complete model paths.

The normal-duration integration validation is complete. The immediate next step is to freeze this checkpoint, then run the authorized owner-instance and regression gates rather than a new baseline comparison. Any missed owner row should drive inspection of recognition, constraints, candidate completion, variable/phase guidance, and scheduling before a new mechanism is invented.